

[Assignment IX]

Problem 1: Projected gradient method. Consider the optimization problem

$$\begin{aligned} & \underset{x \in \mathbb{R}^n}{\text{minimize}} && f(x) + g_C(x) \\ & \text{subject to} && x \in C, \end{aligned}$$

where $C \subset \mathbb{R}^n$. Constrained optimization problems of this type can be solved with the *projected gradient method*

$$x^{k+1} = \Pi_C(x^k - \alpha \nabla f(x^k)),$$

where Π_C is the projection onto C . The projection of $y \in \mathbb{R}^n$ onto $C \subseteq \mathbb{R}^n$ is defined as the point in C that is closest to y :

$$\Pi_C(y) = \underset{x \in C}{\operatorname{argmin}} \|x - y\|^2.$$

For the particular set

$$C = \{x \in \mathbb{R}^2 \mid x_1 = a, 0 \leq x_2 \leq 1\},$$

where $a \in \mathbb{R}$, show that

$$\Pi_C(y) = \begin{bmatrix} a \\ \min\{\max\{y_2, 0\}, 1\} \end{bmatrix},$$

where $y = (y_1, y_2)$.

$$C = \{x \in \mathbb{R}^2 \mid x_1 = a, 0 \leq x_2 \leq 1\}$$

$$\Pi_C(y) = \underset{x \in C}{\operatorname{argmin}} \|x - y\|^2$$

$$= \underset{x_2}{\operatorname{argmin}} \|(y_1, y_2) - (a, x_2)\|^2$$

$$= a, \underset{x_2}{\operatorname{argmin}} \|(y_2 - x_2)\|^2$$

$$\text{if } y_2 < 0 \quad ; \quad x_2 = 0$$

$$y_2 > 1 \quad ; \quad x_2 = 1$$

$$0 \leq y_2 \leq 1 \quad ; \quad x_2 = y_2$$

$$\left. \begin{array}{l} \text{if } y_2 < 0 \quad ; \quad x_2 = 0 \\ y_2 > 1 \quad ; \quad x_2 = 1 \\ 0 \leq y_2 \leq 1 \quad ; \quad x_2 = y_2 \end{array} \right) \Pi_C(y) = \begin{pmatrix} a \\ \min\{\max\{y_2, 0\}, 1\} \end{pmatrix}$$

$$0 \in (A+B)x$$

$$\Leftrightarrow x = \Pi_{\mathbb{R}^n}(I - \alpha A)x$$

Problem 3: VLB for IWAE. The standard variational lower bound (VLB) of VAE is

$$\log(p_\theta(x)) \geq \text{VLB}_{\theta, \phi}(x) = \mathbb{E}_{Z \sim q(Z|x)} \left[\log \left(\frac{p_\theta(x|Z)p_Z(Z)}{q_\phi(Z|x)} \right) \right],$$

where $p_\theta(z|x)$ is the true posterior and $q_\phi(z|x)$ is the approximate posterior. Define

$$\text{VLB}_{\theta, \phi}^{(K)}(x) = \mathbb{E}_{Z_1, \dots, Z_K \sim q_\phi(z|x)} \left[\log \frac{1}{K} \sum_{k=1}^K \frac{p_\theta(x|Z_k)p_Z(Z_k)}{q_\phi(Z_k|x)} \right],$$

to be the VLB for importance weighted autoencoders (IWAE) [1]. To clarify, $Z_1, \dots, Z_K$ are sampled independently from $q_\phi(z|x)$. Note that IWAE with $K=1$ coincides with the standard VAE, and $\text{VLB}_{\theta, \phi}^{(1)} = \text{VLB}_{\theta, \phi}$. Show:

(a) $\log p_\theta(x) \geq \text{VLB}_{\theta, \phi}^{(K)}(x)$ for all x and $K \geq 1$.

$$\begin{aligned} \log p_\theta(x) &= \log \mathbb{E}_{Z \sim p_Z} [p_\theta(x|Z)] = \log \mathbb{E}_{Z \sim q_\phi(Z|x)} \left[p_\theta(x|Z) \frac{p_Z(Z)}{q_\phi(Z|x)} \right] \\ &= \log \mathbb{E}_{Z_1, \dots, Z_K \sim q_\phi(Z|x)} \left[\frac{1}{K} \sum_{k=1}^K \frac{p_\theta(x|Z_k)p_Z(Z_k)}{q_\phi(Z_k|x)} \right] \quad (\because \mathbb{E}: \text{linear}) \\ &\geq \mathbb{E}_{Z_1, \dots, Z_K \sim q_\phi(Z|x)} \left[\log \left[\frac{1}{K} \sum_{k=1}^K \frac{p_\theta(x|Z_k)p_Z(Z_k)}{q_\phi(Z_k|x)} \right] \right] \quad (\because \text{Jensen}) = \text{VLB}_{\theta, \phi}^{(K)}(x) \quad // \end{aligned}$$

(b) If $K \geq M$, then $\text{VLB}_{\theta, \phi}^{(K)}(x) \geq \text{VLB}_{\theta, \phi}^{(M)}(x)$ for all x .

$$\begin{aligned} \text{let } I \subset \{1, 2, \dots, K\}, |I| = M. \\ \mathbb{E}_{Z_1, \dots, Z_M, i \in I} \left[\log \frac{1}{M} \sum_{k=1}^M \frac{p_\theta(x|Z_k)p_Z(Z_k)}{q_\phi(Z_k|x)} \right] &\leq \log \mathbb{E}_{Z_1, \dots, Z_M, i \in I} \left[\frac{1}{M} \sum_{k=1}^M \frac{p_\theta(x|Z_k)p_Z(Z_k)}{q_\phi(Z_k|x)} \right] \\ &= \log \left[\frac{1}{K} \sum_{k=1}^K \frac{p_\theta(x|Z_k)p_Z(Z_k)}{q_\phi(Z_k|x)} \right] \quad \text{for arbitrary } Z_1, \dots, Z_K \sim q_\phi(Z|x) \\ \mathbb{E}_{Z_1, \dots, Z_K} \left[\mathbb{E}_{Z_1, \dots, Z_M, i \in I} \left[\log \frac{1}{M} \sum_{k=1}^M \frac{p_\theta(x|Z_k)p_Z(Z_k)}{q_\phi(Z_k|x)} \right] \right] &\leq \mathbb{E}_{Z_1, \dots, Z_K} \left[\log \frac{1}{K} \sum_{k=1}^K \frac{p_\theta(x|Z_k)p_Z(Z_k)}{q_\phi(Z_k|x)} \right] \\ \text{VLB}_{\theta, \phi}^{(M)}(x) &\leq \text{VLB}_{\theta, \phi}^{(K)}(x) \quad // \end{aligned}$$

(c) Let $X_1, \dots, X_N$ data for training the IWAE. Show that if q_ϕ is “powerful enough”, then

$$\underset{\theta \in \Theta}{\text{maximize}} \sum_{i=1}^N \log p_\theta(X_i) = \underset{\theta \in \Theta, \phi \in \Phi}{\text{maximize}} \sum_{i=1}^N \text{VLB}_{\theta, \phi}^{(K)}(X_i).$$

What should be the precise meaning of “powerful enough”?

If q_ϕ is powerful enough, i.e., $q_\phi(\cdot|x) = p_\theta(\cdot|x)$ & $D_{KL}[q_\phi(\cdot|x) \parallel p_\theta(\cdot|x)] = 0$

$\Rightarrow \log p_\theta(x_i) = \text{VLB}_{\theta, \phi}(x_i) + D_{KL}[q_\phi(\cdot|x_i) \parallel p_\theta(\cdot|x_i)] = \text{VLB}_{\theta, \phi}(x_i)$, and by (b),

$$\sum_{i=1}^N \log p_\theta(X_i) \geq \sum_{i=1}^N \text{VLB}_{\theta, \phi}^{(K)}(X_i) \geq \sum_{i=1}^N \text{VLB}_{\theta, \phi}(X_i) = \sum_{i=1}^N \log p_\theta(X_i) \Rightarrow \underset{\theta \in \Theta}{\text{maximize}} \sum_{i=1}^N \log p_\theta(X_i) = \underset{\theta \in \Theta}{\text{maximize}} \sum_{i=1}^N \text{VLB}_{\theta, \phi}^{(K)}(X_i)$$

Problem 5: Rock paper scissors and minimax optimization. Consider a game of rock paper scissors between players A and B . Players A and B play a randomized strategy with

$$p_A = \begin{bmatrix} \mathbb{P}(A \text{ plays rock}) \\ \mathbb{P}(A \text{ plays paper}) \\ \mathbb{P}(A \text{ plays scissors}) \end{bmatrix}, \quad p_B = \begin{bmatrix} \mathbb{P}(B \text{ plays rock}) \\ \mathbb{P}(B \text{ plays paper}) \\ \mathbb{P}(B \text{ plays scissors}) \end{bmatrix}.$$

Define

$$\Delta^3 = \{p = (p_1, p_2, p_3) \in \mathbb{R}^3 \mid p_1, p_2, p_3 \geq 0, p_1 + p_2 + p_3 = 1\}$$

so that $p_A, p_B \in \Delta^3$. In the game, a player receives 1 point for a win, -1 points for a loss, and 0 points for a draw. Consider the minimax problem

$$\underset{p_A \in \Delta^3}{\text{minimize}} \quad \underset{p_B \in \Delta^3}{\text{maximize}} \quad \mathbb{E}_{p_A, p_B}[\text{points for } B].$$

(a) Show that

$$p_A^* = \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix}, \quad p_B^* = \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix}$$

is the unique solution to the minimax problem.

$$\min_{p_A \in \Delta^3} \max_{p_B \in \Delta^3} \mathbb{E}_{p_A, p_B}[\text{pt}(B)] = \min_{p_A \in \Delta^3} \max_{p_B \in \Delta^3} \{ p_{B,1}(p_{A,3} - p_{A,2}) + p_{B,2}(p_{A,1} - p_{A,3}) + p_{B,3}(p_{A,2} - p_{A,1}) \}$$

If p_A^*, p_B^* is the unique solution, it must satisfy $\nabla_A \mathcal{L}(p_A^*, p_B^*) = 0 = \nabla_B \mathcal{L}(p_A^*, p_B^*)$

$$\Rightarrow p_{A,3} - p_{A,2} = p_{A,1} - p_{A,3} = p_{A,2} - p_{A,1} = 0 \quad , \quad p_A^* \in \Delta^3 \quad \Rightarrow \quad p_{A,1} = p_{A,2} = p_{A,3} = 1/3 \quad (\text{necessary cond.})$$

$$\Rightarrow p_{B,2} - p_{B,3} = p_{B,3} - p_{B,1} = p_{B,1} - p_{B,2} = 0 \quad , \quad p_B^* \in \Delta^3 \quad \Rightarrow \quad p_{B,1} = p_{B,2} = p_{B,3} = 1/3$$

$$\Rightarrow 0 = \mathcal{L}(p_A^*, p_B) \leq 0 = \mathcal{L}(p_A^*, p_B^*) \leq 0 = \mathcal{L}(p_A, p_B^*) \quad : \text{ sufficient } \Rightarrow p_A^*, p_B^* \text{ is unique.}$$

(b) Note that if $p_B = (1/3, 1/3, 1/3)$, then $\mathbb{E}_{p_A, p_B}[\text{points for } B] = 0$ regardless of how A plays. Does this mean any strategy $p_A \in \Delta^3$ is optimal for player A ?

NO, If $p_B = (1/3, 1/3, 1/3)$ but $p_A \neq (1/3, 1/3, 1/3)$, this means that it is not an optimal solution.

If B does not have the value (p_B^*) , it violates the meaning of maximising.